

10 Ans: D

Considering 10.0 g alone, T_2 provides the centripetal force for it.

$$\begin{aligned} T_2 &= m r \omega^2 \\ &= (0.010) (0.150 + 0.050) 6.28^2 = 0.079 \text{ N} \end{aligned}$$

Considering 21.0 g alone, $(T_1 - T_2)$ provides the centripetal force for it.

$$T_1 - T_2 = (0.021) (0.150) 6.28^2 = 0.124 \text{ N}$$

$$T_1 = 0.203 \text{ N}$$

$$\text{Ratio} = 0.203 / 0.079 = 2.6$$

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Anc: B